

PHILADELPHIA INTL, PA, USA

WMO: 724080

Lat: 39.873N

Lon: 75.227W

Elev: 3

StdP: 101.29

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13739

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.1	-7.9	-20.7	0.6	-8.4	-18.3	0.7	-6.2	12.8	1.1	11.8	1.5	5.6	290	0.509

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.0	34.1	23.9	32.7	23.3	31.3	22.5	25.6	31.3	24.8	29.9	24.2	28.7	5.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.0	18.9	28.1	23.3	18.1	27.3	22.7	17.4	26.8	78.5	31.4	75.2	29.8	72.5	28.9	30.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.1	9.5	8.5	-13.0	36.4	3.0	1.6	-15.1	37.5	-16.8	38.5	-18.5	39.4	-20.7	40.6
			-14.3	27.0	2.7	1.2	-16.2	27.8	-17.8	28.5	-19.4	29.1	-21.4	30.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	13.8	1.0	2.3	6.7	12.8	18.1	23.2	26.1	25.1	21.4	14.9	8.8	4.0	(6)
(7)		DBStd	9.70	5.53	5.07	4.96	4.54	4.07	3.42	2.71	2.68	3.49	4.25	4.51	4.78	(7)
(8)		HDD10.0	929	284	219	128	21	0	0	0	0	7	75	196		(8)
(9)		HDD18.3	2450	538	448	363	174	56	4	0	12	123	287	446		(9)
(10)		CDD10.0	2298	4	4	24	106	251	397	498	468	341	158	38	9	(10)
(11)		CDD18.3	779	0	0	1	9	48	151	239	210	103	17	1	0	(11)
(12)		CDH23.3	6787	0	0	8	98	423	1308	2360	1824	673	91	1	0	(12)
(13)		CDH26.7	2484	0	0	1	27	131	473	989	664	186	14	0	0	(13)
(14)	Wind	WSAvg	4.1	4.5	4.6	4.8	4.6	4.0	3.8	3.7	3.4	3.6	3.9	4.1	4.3	(14)
(15)	Precipitation	PrecAvg	1167	95	68	107	95	92	103	115	121	97	92	82	101	(15)
(16)		PrecMax	1502	148	138	195	230	180	238	261	377	265	235	156	228	(16)
(17)		PrecMin	865	41	11	29	29	37	33	41	41	5	15	25	17	(17)
(18)		PrecStd	168	31	32	42	43	32	55	50	68	60	52	34	47	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.0	19.2	24.2	29.9	32.4	34.7	36.4	35.4	33.1	28.4	22.2	19.6	(19)
MCWB			15.1	13.8	15.9	19.0	22.0	23.9	25.4	24.7	22.9	21.0	16.8	15.6	(20)	
2%		DB	15.0	15.6	20.2	26.2	30.1	32.9	34.5	33.3	30.7	25.9	20.0	16.2	(21)	
		MCWB	12.8	11.0	13.6	17.0	20.8	23.1	24.1	23.9	21.9	19.7	15.8	13.4	(22)	
5%		DB	11.6	12.4	17.2	23.2	27.9	31.1	33.0	31.6	28.7	23.7	17.9	13.6	(23)	
		MCWB	9.0	8.3	11.8	15.7	19.8	22.3	23.6	22.9	21.2	18.4	14.2	11.2	(24)	
10%		DB	8.6	9.7	14.5	20.7	25.6	29.5	31.5	30.2	27.1	21.6	15.8	11.0	(25)	
		MCWB	6.0	6.1	10.1	14.1	18.6	21.4	22.9	22.1	20.3	17.2	12.4	8.3	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.0	15.8	17.4	20.4	23.2	25.5	26.8	26.5	24.9	22.8	18.7	16.9	(27)
MCD B			17.3	18.4	22.4	26.7	29.8	31.9	33.4	32.2	29.2	25.6	20.5	18.8	(28)	
2%		WB	13.2	12.0	14.9	18.7	22.1	24.5	25.8	25.5	24.0	21.2	16.9	14.0	(29)	
		MCD B	15.0	14.2	18.4	24.0	28.1	30.1	31.7	30.7	27.7	24.4	19.2	15.8	(30)	
5%		WB	9.3	8.9	12.8	17.1	21.0	23.8	25.0	24.7	23.1	19.7	15.0	11.6	(31)	
		MCD B	11.1	11.5	16.2	21.8	26.1	28.9	30.5	29.1	26.2	22.5	17.2	13.2	(32)	
10%		WB	6.6	6.5	10.4	15.3	19.8	22.9	24.3	23.9	22.1	18.0	13.1	8.9	(33)	
		MCD B	8.2	9.3	14.3	19.5	24.3	27.5	29.1	27.8	25.5	20.9	15.2	10.6	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.3	8.1	9.2	10.2	9.9	9.4	9.0	8.6	8.8	8.7	8.2	7.1	(35)
MCD BR			11.0	12.2	13.5	13.8	12.6	11.2	10.4	9.9	10.3	10.7	10.8	10.2	(36)	
MCWBR			9.3	9.1	8.6	7.1	5.6	4.4	3.6	3.7	4.3	5.7	7.4	9.0	(37)	
5% WB		MCD BR	10.6	11.4	12.8	12.5	11.4	10.1	9.5	8.8	8.5	9.0	9.1	9.8	(38)	
		MCWBR	9.8	9.6	9.4	7.7	5.9	4.7	3.7	3.9	4.7	6.2	7.7	9.4	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.317	0.323	0.366	0.398	0.444	0.507	0.512	0.504	0.416	0.363	0.329	0.312	(40)	
(41)		taud	2.444	2.419	2.357	2.296	2.198	2.066	2.065	2.074	2.294	2.447	2.505	2.509	(41)	
(42)		Ebn at Noon	847	896	886	877	840	785	776	770	823	838	828	828	(42)	
(43)		Edh at Noon	81	95	112	127	144	164	163	157	119	92	76	71	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.94	2.84	3.75	4.85	5.41	5.92	6.02	5.25	4.34	3.11	2.21	1.68	(44)	
(45)		RadStd	0.17	0.31	0.30	0.41	0.65	0.47	0.28	0.33	0.47	0.32	0.20	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.39	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+109	+78	(46)
(47)	Regional (75 neighbors)	+0.42	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+125	+73	(47)

Nomenclature: See separate page